(it reads like this: "The limit of y_n for n tending to infinity is a").

Using the definition of the limit, let us prove that

$$\lim_{n\to\infty} \frac{n}{n+1} = 1 \; ; \; \lim_{n\to\infty} \frac{1}{n} = 0$$

$$\lim_{n\to\infty} \left\{ \frac{1}{2n}\left[1-(-1)^n\right] - \frac{1}{2(n-1)}\left[1+(-1)^n\right] \right\} = 0$$

You will begin with the first problem.

Son: Prove that $\lim_{n\to\infty} \frac{n}{n+1} = 1$

Proof: Take any arbitrary positive number ε_1. We want to show that there exists a real number K where $\left|\frac{K}{K+1} - 1\right| < \varepsilon_1$.

Let, $\left|\frac{K}{K+1} - 1\right| < \varepsilon_1$ hold for some $K \Rightarrow \left|-\frac{1}{K+1}\right| < \varepsilon_1 \Rightarrow \frac{1}{K+1} < \varepsilon_1 \Rightarrow \boxed{\frac{1}{\varepsilon_1} - 1 < K}$

(Wrong argument. You are already assuming what you are trying to prove)

Suppose, $\forall K \in R$, $\left|\frac{K}{K+1} - 1\right| > \varepsilon_1$. $\Rightarrow \left|-\frac{1}{K+1}\right| > \varepsilon_1 \Rightarrow \frac{1}{K+1} > \varepsilon_1$

$\boxed{\Rightarrow \frac{1}{\varepsilon_1} - 1 > K}$. But this is a contradiction. We know that the set of real numbers R has no upper bound. This is because if $x \in R$ is an upper bound on R, then $\forall y \in R$, $y \leq x$. But $x+1 \in R > x$. $\therefore x$ is not an upper bound on R. $\therefore \exists K \in R$ s.t. $\boxed{\left|\frac{K}{K+1} - 1\right| < \varepsilon_1}$.

$\left|\frac{K}{K+1} - 1\right| < \varepsilon_1 \Rightarrow \left|-\frac{1}{K+1}\right| < \varepsilon_1$

$\therefore \forall \varepsilon_1 > 0, \exists K \in R$ s.t. $\boxed{\left|\frac{K}{K+1} - 1\right| < \varepsilon_1}$

$\Rightarrow \frac{1}{K+1} < \varepsilon_1 \Rightarrow \frac{1}{\varepsilon_1} - 1 < K$. $\therefore \forall x \in R$, $x > K$, $\frac{1}{\varepsilon_1} - 1 < x \Rightarrow \frac{1}{x+1} < \varepsilon_1$

$\Rightarrow \left|-\frac{1}{x+1}\right| < \varepsilon_1 \Rightarrow \boxed{\left|\frac{x}{x+1} - 1\right| < \varepsilon_1}$

$\boxed{\therefore \forall \varepsilon_1 > 0, \exists K \in R \text{ s.t. } \forall x \geq K, \left|\frac{x}{x+1} - 1\right| < \varepsilon_1}$ $\square$

Reader: I have to prove that $\lim_{n\to\infty} \frac{n}{n+1} = 1$. I choose an arbitrary value of ε, for example, $\varepsilon = 0.1$.

Author: I advise you to begin with finding the modulus of $|y_n - a|$.

Reader: In this case, the modulus is $\left|\frac{n}{n+1} - 1\right| = \frac{1}{n+1}$